

Non-proportionality of Abel–Prym cycles for cyclic étale triple covers

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Abstract

Let $\pi: \tilde{C} \rightarrow C$ be an étale cyclic $\mathbb{Z}/3$ cover of a smooth projective curve of genus $g \geq 2$, with Prym variety B of dimension $2n$, $n = g - 1$, and induced polarization Θ . We prove that the middle Abel–Prym cycle class $W_n = (\varphi_n)_*[\text{Sym}^n \tilde{C}]$ is never proportional to Θ^n ; equivalently, $\Xi_n = W_n - \Theta^n/(n!3^n)$ is a nonzero, Θ -primitive, algebraic class in the middle cohomology of B , in every dimension at once. This contrasts sharply with the classical étale double-cover case, where Abel–Prym cycles represent twice the minimal class and in particular lie on the theta line (Welters, Beauville). The proof is an integrality argument: the intersection numbers $W_k \cdot \Theta^{2n-k} = k! \binom{2n}{k}$ follow from an exact lattice-index identity $\text{Pf}(sE + tE_V) = s^g(s+t)^{2(g-1)}$, the induced polarization has type $(1^n, 3^n)$, and proportionality would force $W_n^2 = \binom{2n}{n}/3^n$, contradicting integrality of the self-intersection: the 3-adic valuation of the central binomial coefficient equals the number of carries in the base-3 addition $n + n$ and is therefore always less than n . The lattice inputs are verified by exact machine computation on integral models, and the arithmetic core is formalized in Lean 4 with the dimension as a bound variable.

1 Introduction

For an étale double cover, the geometry of the Abel–Prym embedding is governed by a classical fact: the Abel–Prym curve represents *twice the minimal class* $2\theta^{p-1}/(p-1)!$ on the principally polarized Prym (P, θ) , and Beauville’s special subvarieties provide representatives of twice the minimal class in every intermediate dimension [1, 2]. In particular all these cycles lie on the theta line in rational cohomology: the double-cover Abel–Prym geometry produces no primitive algebraic classes.

The purpose of this note is to prove that the cyclic triple-cover situation is rigidly different, in every dimension simultaneously.

Theorem 1. *Let $\pi: \tilde{C} \rightarrow C$ be an irreducible étale cyclic $\mathbb{Z}/3$ covering of a smooth projective complex curve of genus $g \geq 2$. Let $B = \ker(\text{Nm})^0 \subset J\tilde{C}$ be the Prym variety, $\dim B = 2n$ with $n = g - 1$, and let Θ denote the polarization induced by the canonical principal polarization of $J\tilde{C}$. Let $\varphi_n: \text{Sym}^n \tilde{C} \rightarrow B$ be an Abel–Prym map and $W_n = (\varphi_n)_*[\text{Sym}^n \tilde{C}] \in H^{2n}(B, \mathbb{Q})$. Then $W_n \notin \mathbb{Q} \cdot \Theta^n$. Equivalently,*

$$\Xi_n = W_n - \frac{1}{n!3^n} \Theta^n$$

is a nonzero algebraic class with $\Xi_n \cdot \Theta^n = 0$.

The proof has three inputs, each elementary once isolated.

(i) *The battery.* Writing E for the principal symplectic form of $H_1(J\tilde{C}, \mathbb{Z})$ and E_V for its restriction-projector to the Prym part, an exact lattice-index computation gives the two-variable Pfaffian identity

$$\text{Pf}_\Lambda(sE + tE_V) = s^g (s + t)^{2(g-1)},$$

from which, by the Poincaré formula on $J\tilde{C}$ and the projection bookkeeping of the norm map, the intersection numbers $q_k := W_k \cdot \Theta^{2n-k} = k! \binom{2n}{k}$ follow; in particular $q_n = n! \binom{2n}{n}$ and $\Theta^{2n} = (2n)! 3^n$.

(ii) *The type.* The induced polarization on the $\mathbb{Z}/3$ -Prym has type $(1^n, 3^n)$ (Lange–Ortega [3]; verified independently by Smith reduction on the integral model below), so $\text{Pf}(E|_B) = 3^n$.

(iii) *The trap.* If $W_n = \lambda \Theta^n$ then $\lambda = q_n / \Theta^{2n} = 1 / (n! 3^n)$ and

$$W_n^2 = \lambda^2 \Theta^{2n} = \frac{(n! \binom{2n}{n})^2}{(2n)! 3^n} = \frac{\binom{2n}{n}}{3^n},$$

using $(n! \binom{2n}{n})^2 = \binom{2n}{n} (2n)!$. But W_n is the class of an integral cycle, so $W_n^2 \in \mathbb{Z}$, while by Kummer’s theorem $v_3(\binom{2n}{n})$ equals the number of carries in the base-3 addition of $n + n$, hence is at most $\lfloor \log_3 2n \rfloor < n$ for all $n \geq 1$. Contradiction. $\square$

Remark 2 (Contrast with $\mathbb{Z}/2$). *For double covers the analogous computation lands exactly on the integer: the Abel–Prym classes are the (twice-)minimal classes, and no obstruction occurs — the theta-proportionality of [2, 1] and the non-proportionality above are two values of the same lattice computation, distinguished by the glue exponent of the induced polarization. The μ_3 -harmonic rigidifies the cycle.*

Remark 3 (Verification register). *The identity in (i) and the type in (ii) are verified by exact integer linear algebra on the topological cut-and-glue model of the cover (n sheets of the cut base surface; the model is unique by mapping-class transitivity on primitive $\mathbb{Z}/3$ -classes): the symbolic Pfaffian $\text{Pf}(sE + tE_V)$ and the Smith normal form of $E|_B$ are computed exactly at $g = 4, 5$ (`tmp/integral_model.sage`, `tmp/integral_model_g5.sage`), the model itself is validated against Lange–Ortega at both $n = 3$ and $n = 6$, and the handle-block structure gives the general g . The arithmetic of (iii) is formalized in Lean 4 with n a bound variable (`UniversalTrap.lean: centralBinom_three_adic_lt, universal_integrality_obstruction, PrymTrapData.abel_prym_rigidity`; axiom footprint `{propext, Classical.choice, Quot.sound}`). The classical inputs are the Poincaré formula and integrality of self-intersections of cycle classes. What is proved is cited by Lean name or script name, and nothing else is claimed. This note is the standalone elaboration of the paragraph “The universal rigidity law” of the companion paper [5], whose registers and falsification record it inherits: the trap identity was found while pursuing — and retracting, against the literature — a stronger claim, and that retraction is part of this note’s provenance.*

Remark 4 (Relation to algebraicity of Weil classes). *On these Pryms the two-dimensional space of Weil Hodge classes is known to be algebraic (Schoen; Patel–Zhang [4] for general abelian covers). Theorem 1 is complementary content: it does not produce the Weil classes,*

but locates the Abel–Prym cycle itself off the theta line — equivalently, it exhibits an explicit nonzero Θ -primitive algebraic class with an explicit cycle representative on every such Prym, including on members where the Hodge ring is as small as the Weil-type structure allows. For simple members the statement has independent geometric force; explicit simple examples in dimensions 6 and 8, certified by irreducible Frobenius polynomials at split primes and exact purity, accompany this note in the repository, together with the exact integral-model computations.

References

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